

K.R MANAGALAM UNIVERSITY



Project Report

Learning Management System (LMS)

Project Title – LearnSphere

Web Dev Capstone Project

Submitted by – Anupriya Agarwal

Roll no. – 2501420005

1. Introduction

The Learning Management System (LMS) is a full-stack web application designed to provide an interactive and modern platform for online learning. The system allows students to browse courses, enroll, watch video lectures, and track their progress. Additionally, instructors or administrators can manage course content efficiently. With the rapid growth of digital education, this platform aims to replicate and enhance the experience offered by platforms like Udemy and Coursera, while maintaining simplicity and performance.

2. Objectives

- To build a scalable and high-performance LMS web application
- To provide seamless course browsing and enrollment experience
- To enable video-based learning with progress tracking
- To implement secure authentication using JWT
- To create separate dashboards for students and administrators

3. System Architecture

The application follows a monorepo architecture with two main components:
Frontend

- React (Vite) for fast development and rendering
- Tailwind CSS for styling and responsive design
- Redux Toolkit for state management
- React Router for navigation

Backend

- Node.js with Express.js
- MongoDB with Mongoose ORM
- JWT-based authentication

Architecture Flow

1. User interacts with frontend UI
2. Frontend sends API requests via Axios
3. Backend processes requests and interacts with database
4. Data is returned and rendered dynamically

4. Methodology

The project follows a System Development Life Cycle (SDLC) approach:

4.1 Requirement Analysis

- Identify user roles: Student and Admin
- Define system functionalities such as authentication, course management, and progress tracking

4.2 Design Phase

- UI/UX designed using Tailwind CSS
- Database schema created for Users, Courses, and Lessons

4.3 Development Phase

- Backend APIs developed using Express
- Frontend components built using React
- Integration done via REST APIs

4.4 Testing Phase

- API testing using Postman
- UI testing using browser tools

4.5 Deployment (Optional)

- Frontend can be deployed on Vercel
- Backend can be deployed on Render or Railway

5. Key Features

5.1 Authentication

- User registration and login
- JWT-based secure authentication

5.2 Course Management

- Create, update, delete courses (Admin)
- Browse and enroll in courses (Student)

5.3 Learning Experience

- Video player integration (YouTube or custom)

- Course modules and lessons
- Progress tracking system

5.4 Dashboards

- Student dashboard with enrolled courses
- Admin dashboard with analytics and course management

5.5 UI/UX Features

- Dark mode toggle
- Responsive design
- Smooth animations

6. Database Design

Collections:

User

- Name
- Email
- Password
- Role (Student/Admin)
- Enrolled Courses

Course

- Title
- Description
- Category
- Thumbnail
- Modules

Lesson/Module

- Title
- Video URL
- Notes

7. Implementation Details

Backend APIs

- /api/auth/register – Register user
- /api/auth/login – Login user
- /api/courses – CRUD operations for courses
- /api/progress – Track learning progress

Frontend Modules

- Authentication pages (Login/Signup)
- Course listing page
- Course detail page
- Video learning interface
- Dashboards

8. Tools & Technologies

Category	Technology
Frontend	React, Vite, Tailwind CSS
Backend	Node.js, Express.js
Database	MongoDB
State Management	Redux Toolkit
API Testing	Postman
Deployment	Vercel, Render

9. Testing & Verification

Automated Testing

- API endpoint validation using Postman
- Checking Redux state updates

Manual Testing

- User registration and login flow
- Course enrollment and progress tracking
- Admin course creation and updates
- Responsive design testing

10. Results & Outcomes

- Successfully developed a full-stack LMS platform
- Implemented real-time course tracking system
- Achieved responsive and user-friendly UI
- Ensured secure authentication system

11. Advantages

- Easy access to learning materials
- Scalable architecture
- User-friendly interface
- Role-based access control

12. Limitations

- No live video streaming (only recorded videos)
- Limited analytics features
- No payment integration (can be added in future)

13. Future Enhancements

- Add payment gateway integration
- Implement live classes and chat system
- Add AI-based course recommendations
- Mobile app version

- Advanced analytics dashboard

14. Conclusion

The LMS project successfully demonstrates the implementation of a modern full-stack web application using industry-standard technologies. It provides a strong foundation for scalable e-learning platforms and can be further enhanced with advanced features like AI recommendations, real-time collaboration, and monetization.

15. References

- [MongoDB Documentation](#)
- [React Documentation](#)
- [Express.js Documentation](#)
- [Tailwind CSS Documentation](#)
- [JWT Authentication Guide](#)